



Cairo University
 Faculty of Engineering
 Credit Hours System
 Communications and Computer Engineering Department

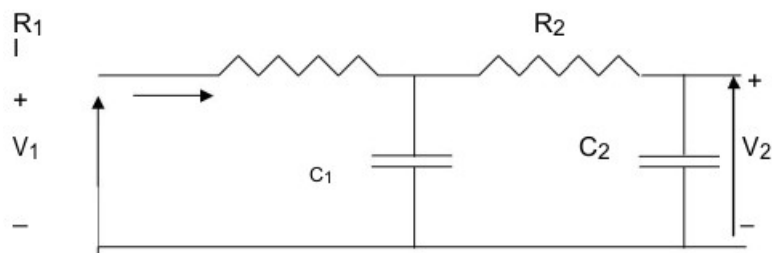
Fall 2026

Course Code EECS 304
 Course Title Control I
 Course Instructors Dr. Ragia Badr - Dr. Nasr Antar
 Course TA Eng. Hassan El-Menier

Sheet 2: Transfer Function

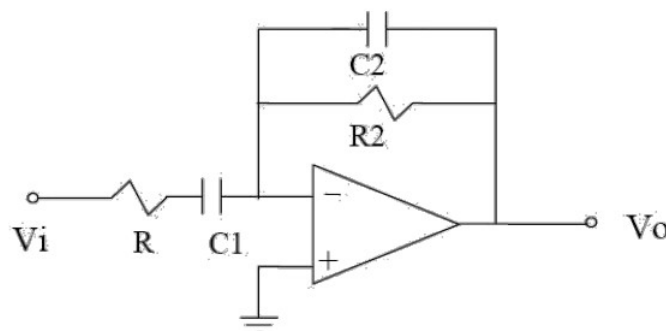
Question (1):

For the circuit shown, obtain the transfer function $V_2(s)/V_1(s)$.



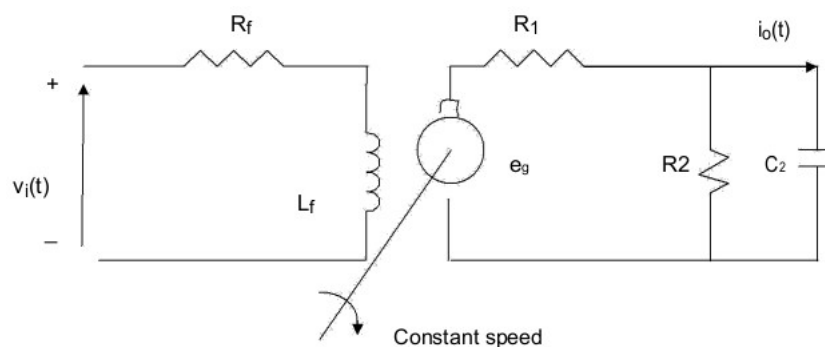
Question (2):

For the ideal OP Amp, obtain the transfer function $V_o(s)/V_i(s)$.



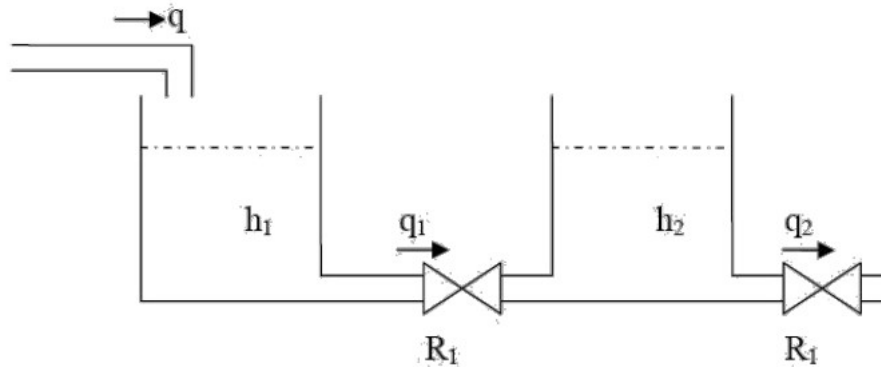
Question (3):

The system shown consists of a DC generator connected to an RC impedance. Find the TF that relates V_i and I_o .



Question (4):

The figure shows a process plant containing two tanks of areas A_1 and A_2 . Derive the transfer function that relates q_2 to q .

**Question (5):**

Derive the differential equations that represent the mechanical system shown, where U is a force that affects the mass M_1 and hence derive the transfer functions $X_1(s)/U(s)$ and $X_2(s)/U(s)$. If the force U has a value of 1 N, find the steady state values of X_1 and X_2 .

